CANDIDATE'S NAME:…….....

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Index number							

535/1

Physics

Paper 1

2¼ hours

Apex Examination Council

PRE-ENTRY EXAMINATIONS SET TWO 2026

Uganda certificate of education

PHYSICS

Paper 1

2 hours 15 minutes

INSTRUCTIONS TO STUDENTS:

This paper consists of two sections; A and B. It has seven examinations items.

Section A has three compulsory items.

Section B has two parts; I and II. Answer one item from each part.

Answer five items in all.

Answers to all items must be written on the answer sheet(s) provided.

SECTION A

Attempt all questions

Item 1

In Uganda, total darkness was experienced during day time in Pakwachi though many people were left surprised what could cause that, though some were saying the "moon was fighting with the sun". The planning authority in Kyle's area passed new guideline requiring that all perimeter walls less than 10m away from the road sides must be broken down. Kyle's perimeter wall near the road seems to be in danger of being broken down. Kyle lacked any measuring instrument to establish the actual distance of that wall from the rod side. Kyle's neighbour across the road advised him to use echo method. His neighbour's wall is 60m from the side of the road opposite that of Kyle's residence. The road is 8m wide Kyle stood alongside his perimeter wall that is close to the road and clapped two pieces of wood, once. He heard the reflected sound from the neighbour's wall 0.5 seconds later. However, Kyle and his neighbour did not know how to sue information obtained.

Of late, some young men have been consistently converging along the road near Kyle's perimeter wall. He wants to be watching them closely without the young men seeing him, but Kyle lacks installed security camera at his residence.

Kyle has contacted you for help.

<u>Hint</u>

- The speed of sound in air is $330ms^{-1}$
- Kyle's perimeter walls near the road, the road and the perimeter wall of the neighbour across the road are parallel to each other.

Task

a) Explain the process that led to total darkness
b) Determine whether his perimeter wall will be among those to be broken down.

Item 2

The government has identified land in a certain area for establishing a nuclear power plant. As a standard procedure, the government has informed the people living in that area that they have to be relocated, since they cannot co-exist with the nuclear power plant. However, majority of the people have insisted that they cannot leave their land and think they can co-exist with the nuclear power plant. The government has promised that after a specific period of time the community members will be allowed to return.

Due to this, data about the activity of the radioactive materials to be used in the nuclear power plant has been provided to the people in table I.

Table I

Count rate (per minute)	800	525	335	215	150
Time ($1x10^9$years)	0	4	8	12	18

Unfortunately, none of the community members can make sense of the data.

The community members also want to know if there are any benefits of radioactive materials in general.

You have been contacted by the community members for help.

Hint; The environment would be safe from waste radiation when the count rate falls below 35 counts per minute.

Task

a) Help the community members to;
   i) Understand the need for them to be relocated as soon as possible
   ii) Determine the time that has to pass before they can return to settle around the power plant again. (Use the data in table I)
b) Describe any two benefits the community members are interested in.
c) Describe the operation of a nuclear reactor.

Item 3

During dry season, a sailor in an ocean embarked on a journey to transport fish to a certain island for packaging. The sailor was not sure of the direction to take in order to reach the destination (island) safely without getting lost. The sailor contacted her friend who was already at the destination. In a bid to help the sailor, her friend sent google map link and guided the sailor on how to use it. The sailor followed it up to reach the destination (island) without getting lost. However, the sailor did not understand how the google map works.

On arrival at the destination (island), the sailor noticed that the level of water started rising in the ocean at night: where the waves carried large volumes. This could cause any sailor's boat to become un stable and eventually capsize. The sailor wondered whether there are benefits of such rising water levels, to the communities living around such water bodies during dry season.

Task

a) Help the sailor to understand
   (i) Why water behaved the way it did.
   (ii) How the application sent by her friend works?
b) Explain any two behaviour observed
c) Explain how a live broadcast is abled for one to watch another on TV from different areas.

SECTION B

PART I- (Attempt one question)

Item 4

Students from a certain school had a trip to the hydroelectric power plant. The person in charge at the plant took them around. They at first went to see the wall of the dam. The person in charge told the students that the wall of such a dam is always built thicker and under at the bottom than at the top. The students did not know why the wall of the dam should be built that way and feared to ask the in charge, yet they were eager to know.

Later, the person in charge took the students to a nearby construction site. At the site, a mixture of sand, gravel, cement, water and steel bars was being made to erect columns for a storeyed building. The person in charge told the students that each of the steel bars being used are of length 3.2m, diameter of 2.5cm, and would stretch by 0.2mm when a tensile force of $7.7x10^5$N is applied. The students wondered whether the steel bars met the requirements for use in the structure, given the heavy weight of the building. After lunch, the student visited a beach and noticed that while standing out of water, they felt a cool breeze compared to when they were sailing in a boat and wondered why the difference. One of the students quickly mentioned that there could be a number of applications of the breeze they experienced, in real life. Majority of the students did not now any practical applications of the breeze they experienced, since they had not yet learnt but were interested in finding out.

On returning from the trip, the students approached you for help.

Hint: Steel bars are of good quality if the ratio of stress to strain exceeds $2.1x10"Nm^{-2}$.

Task

a) Explain the students why the wall they observed was designed as they observed.
b) Help the students to determine whether steel bars were of good quality.
c) Explain to the students any two practical applications of the breeze they experienced.

Item 5

Mugala hospital, they use an instrument to measure temperature of sick people, but some people don't know how they are used. These instruments have liquids in them. The sick people, example women who go to give birth go with an instrument which keeps hot things hot and cold things cold. However, when hot water is put in a plastic cup and in a metallic cup at the same time, after a while the one in a metallic cup cools faster than that in a plastic cup. In the morning when I touched the metal bars, they felt so cold and that puzzled me too. When I went to school, my teacher tasked me to convert 243K to degrees centigrade and 48^0C to kelvin. But I failed the work.

Task

i) Explain the steps to take when using the instrument to measure the temperature.
ii) With illustration, explain how the instruments keep hot things hot and cold things cold.

iii) Explain the process that took place when hot water was placed in plastic and metallic cup at the same time.
iv) Why did he feel cold after touching the bar in the morning?
v) Help him solve the work given to him at school.
vi) What are qualities of a good thermometric liquid.

PART II- Attempt one question

Item 6

A business person owns a company that uses current electricity to operate machines and other appliances. One day, as the workers in the company were switching on the lights, they noticed that all the bulbs were dim. Subsequently, the business person called an electrician to assess the problem who later established that the mains supply of 415V was low and needed to be increased to 2000V. The electrician assured the business person that for the problem to be solved, the business person must buy an efficient device such that the power loss does not exceed 5W. The business person does not have any information on practical ways of how to minimise energy losses in the device.

You have been contacted by the business person for help.

Hint; Cables of resistance 2Ω supply 2KW of electrical power, if it is transmitted at 2000V.

Task

a) Help the business man understand the design and mode of operation of the device needed to sort out the voltage issue.
b) Describe to the business person any two practical ways of minimising energy losses in the device.

Item 7

A business person plans to set up a new restaurant which has only one socket for the mains. The business person wants to connect the items shown in table 2 for use.

Table 2

S/N	Item	Power rating
1	Loudspeaker	240V, 60W
2	Electric kettle	240V, 1KW
3	Torch	12V, d.c
4	T.V	12V, d.c

However, the business person does not know how they can be connected. In order to budget properly, the business person wants to know the cost of electrical energy needed for operating a loud speaker for 10 hours a day per week. The business person was told that it is

possible to pay less than the expected amount if practical steps are taken to save energy. However, the idea of saving energy sounded new to the business person and wanted to be educated about it.

The business person was also told that there are some missing electrical components needed to be connected with the items in table 2. It was not easy to tell the missing components.

You have been contacted by the business person to offer guidance.

Hint. The unit cost of electrical energy is shs 600.

Task:

a) Help the business person to;
i) Identify the missing components required for the items in table 2 to be connected safely.
ii) Explain the purpose of each component
iii) Illustrate, using a circuit diagram, how the devices in table 2 can be connected with the identified components.
b) Find out how much money is needed to operate the loudspeaker as planned, so that the business person can budget accordingly.
c) Explain any two practical steps the business person can take, in order to minimise energy losses to save energy and pay less for energy consumed.

END

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